

Date	03 July 2026
Team ID	
Team Size	3
Team Leader	Chakali Basavaraju
Team Member 1	Girish C
Team Member 2	Shaik Sameer
Team Member 3	
Team Member 4	
Project Name	Smart Lender: Machine Learning-Based Loan Approval Prediction System
Maximum Marks	

Code Layout, Readability, and Reusability

This document evaluates the quality of the codebase for the Smart Lender Loan Approval Prediction System in terms of structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable, scalable, and can be extended or reused in future development.

1. Code Quality Parameters

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	PEP 8 standards followed
2	Proper File Structure	Files and folders are logically organized	Yes	Clear separation of modules, templates, static files, and models
3	Meaningful Variable Names	Variables clearly reflect their purpose	Yes	Descriptive and readable variable names used
4	Function / Method Names	Functions are descriptively named	Yes	Functions are named based on their responsibility
5	Code Comments	Inline and block comments explain logic	Yes	Important logic is well documented
6	Modular Design	Code is divided into reusable modules/functions	Yes	Reusable functions and modular scripts implemented
7	No Redundant Code	Duplicate or unused code is removed	Yes	No redundant or unnecessary code found

2. Component Reusability

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Data Preprocessing Module (preprocess.py)	Python, Pandas, Scikit-learn	Used in training and prediction modules	High
2	Prediction Module (predict.py)	Python, Scikit-learn	Used for model inference and prediction generation	High

3	Database Module	Python, MySQL	Used for authentication, loan records, and prediction history	High
4	Frontend Templates	HTML5, CSS3, Bootstrap, JavaScript	Reused across Login, Registration, Dashboard, and Prediction pages	High
5	Machine Learning Model	Scikit-learn, Joblib	Used by prediction engine to classify loan approval status	High

3. Code Layout Checklist

Aspect	Rating (1–5)	Comments
Code Layout & Structure	5	Well-organized project structure with clear separation of frontend, backend, database, and ML components
Reusability	5	Reusable modules for preprocessing, prediction, authentication, and database operations
Documentation / Comments	5	Source code contains appropriate comments and documentation for better understanding

4. Overall Code Quality Assessment

The Smart Lender Loan Approval Prediction System demonstrates excellent coding standards and software engineering practices. The codebase is clean, modular, and easy to understand, making it highly maintainable and scalable for future enhancements.

Overall Score

5 / 5

The project achieves a high level of code quality with strong emphasis on readability, modularity, maintainability, and reusability, adhering to best practices in software development.

